

Ioana Oriana Bercea

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Positions

- 1/10/2023–
present **Assistant Professor (tenure-track),**
DIVISION OF THEORETICAL COMPUTER SCIENCE
Department of Computer Science, School of Electrical Engineering and Computer Science
KTH ROYAL INSTITUTE OF TECHNOLOGY, Stockholm, Sweden
- 1/05/2021–
30/09/2023 **Postdoc,**
BASIC ALGORITHMS RESEARCH COPENHAGEN (BARC)
Department of Computer Science
IT UNIVERSITY OF COPENHAGEN, Copenhagen, Denmark
Host: Thore Husfeldt
- 1/09/2018–
30/04/2021 **Postdoc,**
School of Electrical Engineering
TEL AVIV UNIVERSITY, Tel Aviv, Israel
Host: Guy Even

Education

- 03/09/2013–
17/08/2018 **Ph.D. in Computer Science,**
UNIVERSITY OF MARYLAND, College Park, MD, USA
Thesis: *Approximation Algorithms for Geometric Clustering and Touring Problems*
Advisor: Samir Khuller
- 30/08/2010–
19/05/2013 **M.Sc. in Computer Science,**
UNIVERSITY OF MARYLAND, College Park, MD, USA
Advisor: Aravind Srinivasan
- 25/09/2006–
12/06/2010 **B.Sc. in Mathematics (Honors) and B.Sc. in Computer Science,**
UNIVERSITY OF CHICAGO, Chicago, IL, USA
Four-year scholarship for tuition, housing and stipend (unconditioned on GPA)

Honors and Awards

- 2026 **Best Research Paper Honorable Mention, SIGMOD**
- 2026 **Research Highlight, SIGMOD**
- 2025 **Best Research Paper, VLDB**
- 2023 **Best Artifact, ACM SIGMOD**
- 2023 **Future Digileader, DIGITAL FUTURES**
- 2014–2015 **Outstanding Graduate Assistant Award, UNIVERSITY OF MARYLAND**
Top 2% of all UMD Graduate Assistants
- 2010–2012 **Dean's Fellowship, UNIVERSITY OF MARYLAND**
- 2006–2008, 2009–2010 **Dean's List, UNIVERSITY OF CHICAGO**

Grants

- 2025 – 2028 **Starting Grant, SWEDISH RESEARCH COUNCIL**
Natural and engineering sciences, 1 100 000 SEK/year

Refereed Conference Articles (Author order alphabetical, unless stated otherwise)

- ESA'26 **Learning-Augmented Online Sorting and TSP**
Ioana-Oriana Bercea, Gerth Stølting Brodal, John Iacono, László Kozma, Debmalya Panigrahi
In: *European Symposium of Algorithms*, (ESA)
- ICML'26 **Instance-Specific Approximation Ratios for Correlation Clustering and Max-Cut**
Sebastian Lüderssen, Ioana-Oriana Bercea, Stefan Neumannn (authors in order of contribution)
In: *International Conference on Machine Learning*, (ICML)
- SIGMOD'26 **Sublime: Sublinear Error & Space for Unbounded Skewed Streams**
Navid Eslami, Ioana O. Bercea, Rasmus Pagh, Niv Dayan (authors in order of contribution)
In: *International Conference on Management of Data*, (SIGMOD)
Best Research Paper Honorable Mention
- VLDB'25 **Diva: Dynamic Range Filter for Var-Length Keys and Queries**
Navid Eslami, Ioana O. Bercea, Niv Dayan (authors in order of contribution)
In: *Very Large Data Bases Conference*, (VLDB)
Best Research Paper
SIGMOD 2026 Research Highlight
- ICALP'25 **Algorithms for the Diverse-k-SAT problem: the geometry of satisfying assignments**
Per Austrin, Ioana O. Bercea, Mayank Goswami, Nutan Limaye, and Adarsh Srinivasan
In: *EATCS International Colloquium on Automata, Languages, and Programming*, (ICALP)
Invited talk to Highlights of Algorithms (HALG) 2026
- CIAC'25 **Dynamic Filter and Retrieval with One Access to Modifiable Memory**
Ioana O. Bercea, Guy Even, Tomer Even and Gabriel Marques Domingues
In: *International Conference on Algorithms and Complexity*, (CIAC)
- ESA'24 **Online sorting and online TSP: randomized, stochastic, and high-dimensional**
Mikkel Abrahamsen, Ioana O. Bercea, Lorenzo Beretta, Jonas Klausen, László Kozma
In: *European Symposium on Algorithms*, (ESA)
- VLDB'24 **Aleph Filter: To Infinity in Constant Time**
Niv Dayan, Ioana O. Bercea, Rasmus Pagh (authors in order of contribution) (authors in order of contribution)
In: *Very Large Data Bases Conference*, (VLDB)
- SWAT'24 **Daisy Bloom Filters**
Ioana O. Bercea, Jakob Bæk Tejs Houen, Rasmus Pagh
In: *19th Scandinavian Symposium and Workshops on Algorithm Theory*, (SWAT)
- FOCS'23 **Locally Uniform Hashing**
Ioana O. Bercea, Lorenzo Beretta, Jonas Klausen, Jakob Bæk Tejs Houen, Mikkel Thorup
In: *64th IEEE Symposium on Foundations of Computer Science*, (FOCS)
- SIGMOD'23 **InfiniFilter: Expanding Filters to Infinity and Beyond**
Niv Dayan, Ioana O. Bercea, Pedro Reviriego, Rasmus Pagh (authors in order of contribution)
In: *International Conference on Management of Data*, (SIGMOD)
Best Artifact
- STOC'22 **An Extendable Data Structure for Incremental Stable Perfect Hashing**
Ioana O. Bercea, Guy Even
In: *54th Annual ACM Symposium on Theory of Computing*, (STOC)
- WADS'21 **Dynamic Dictionaries for Multisets and Counting Filters with Constant Time Operations**
Ioana O. Bercea, Guy Even
In: *17th Algorithms and Data Structures Symposium*, (WADS)
Invited to the special issue of Algorithmica

- CIAC'21 **Upper Tail Analysis of Bucket Sort and Random Tries**
Ioana O. Bercea, Guy Even
In: *12th International Conference on Algorithms and Complexity*, (CIAC)
Invited to the special issue of Theoretical Computer Science
- SWAT'20 **A Dynamic Space-Efficient Filter with Constant Time Operations**
Ioana O. Bercea, Guy Even
In: *17th Scandinavian Symposium and Workshops on Algorithm Theory*, (SWAT), pp. 11:1-11:17
- APPROX'19 **On the cost of essentially fair clusterings**
Ioana O. Bercea, Martin Groß, Samir Khuller, Aounon Kumar, Clemens Rösner, Daniel R. Schmidt, Melanie Schmidt
In: *22nd International Workshop on Approximation Algorithms for Combinatorial Optimization Problems*, (APPROX), pp. 18:1-18:22
- CCCG'18 **Improved Bounds for the Traveling Salesman Problem with Neighborhoods on Uniform Disks**
Ioana O. Bercea
In: *30th Canadian Conference on Computational Geometry*, (CCCG), pp.129-141
- CCCG'16 **Minimizing Uncertainty through Sensor Placement with Angle Constraints**
Ioana O. Bercea, Volkan Isler, Samir Khuller
In: *28th Canadian Conference on Computational Geometry*, (CCCG), pp. 287-294
- SPAA'14 **On Computing Maximal Independent Sets of Hypergraphs in Parallel**
Ioana O. Bercea, Navin Goyal, David G. Harris, Aravind Srinivasan
In: *26th ACM Symposium on Parallelism in Algorithms and Architectures*, (SPAA), pp. 42-50
Invited to the special issue of TOPC

Journal Articles (Author order alphabetical, unless stated otherwise)

- 2026 **Dynamic Range Filtering Beyond Worst-Case Bounds**
Navid Eslami, Ioana O. Bercea, Niv Dayan *(authors in order of contribution)*
In: *ACM SIGMOD Record*, Volume 55, Issue 1
- 2022 **Dynamic Dictionaries for Multisets and Counting Filters with Constant Time Operations**
Ioana O. Bercea, Guy Even
In: *Algorithmica*, Special issue on WADS 2021
- 2021 **Upper Tail Analysis of Bucket Sort and Random Tries**
Ioana O. Bercea, Guy Even
In: *Theoretical Computer Science*, Special issue on CIAC 2021
- 2016 **On Computing Maximal Independent Sets of Hypergraphs in Parallel**
Ioana O. Bercea, Navin Goyal, David G. Harris, Aravind Srinivasan
In: *ACM Transactions on Parallel Computing*, Special issue on SPAA 2014

Invited Workshops

- Feb. 2026 NII SHONAN: *New Era of Data Structures: From Strings to Graphs, and more*
- Aug. 2025 DAGSTUHL : *Software Performance Engineering*
- May 2025 DAGSTUHL : *Adaptive and Scalable Data Structures*
- May 2025 DAGSTUHL : *Learned Predictions for Data Structures and Running Time*
- March 2025 NII SHONAN: *Algorithmic tools and their applications in emerging models of computation*
- July 2024 NU SINGAPORE: *Algorithms and Data Structures Today*
- Sept 2023 BERTINORO : *Algorithmic Aspects of Clustering and Related Problems*
- May 2023 DAGSTUHL : *Scalable Data Structures*
- March 2023 NII SHONAN : *New Frontiers in Locality in Computation*
- Feb. 2021 DAGSTUHL : *Scalable Data Structures*

Oct. 2019 IGAFIT: *Workshop for Postdoctoral Researchers in Algorithms*
Sept. 2019 GI-DAGSTUHL : *Algorithms for Big Data*
March 2019 DAGSTUHL : *Theoretical Foundations of Storage Systems*
Jan. 2019 DAGSTUHL: *Data Structures for the Cloud and External Memory Data*
Oct. 2018 7th French-Israeli Workshop on Foundations of Computer Science

Invited Talks

June 2026 RANDOMNESS IN COMPUTING, University of Sheffield, UK
Dec 2025 A CELEBRATION OF TCS, Sydney, Australia
Oct 2025 WAVE (WOMEN IN ALGORITHMS, VENTURE INTO EXPLORATION), Copenhagen, Denmark
Aug. 2025 TU WIEN
June 2025 BOCCONI THEORY DAY, Milan, Italy
March 2025 TOKYO TECH
Dec 2024 AUSTRALASIAN SUMMER SCHOOL ON RECENT TRENDS IN ALGORITHMS, Sydney, Australia
July 2023 ROMANIAN ALGORITHMS DAYS, virtual

PhD supervision

2026 – **Fabien Niessen**
KTH ROYAL INSTITUTE OF TECHNOLOGY
primary supervisor: Aris Gionis
2025 – **Riccardo Zuech**
UNIVERSITY OF PADOVA
co-supervised with Francesco Silvestri
2021-2025 **Jonas Østergaard Klausen**
UNIVERSITY OF COPENHAGEN
co-supervised with Mikkel Thorup, Jacob Holm
→ postdoc at MPI Saarbrücken

Teaching

Fall 2023 – **Course Responsible, Teacher for DD2440 Advanced Algorithms, MsC,**
KTH ROYAL INSTITUTE OF TECHNOLOGY, Department of Computer Science
Summer 2022 **Co-Instructor for Algorithms and Data Structures BSc, MSc,**
& 2023 IT UNIVERSITY OF COPENHAGEN, Department of Computer Science
Spring 2016 **Co-Instructor for CMSC 122: Intro to Computer Programming via the Web,**
UNIVERSITY OF MARYLAND, Department of Computer Science

Service

2028 **Dagstuhl seminar co-organizer** *Simplicity and Efficiency of Data Structures (28221)*
2026 **Conference organizer** *Highlights of Algorithms (HALG 2026), A celebration of complexity*
2025 **Program Committee co-chair** SOSA 2025 (with Rasmus Pagh)
Program Committee member
2026 ESA (TRACK S), EUROPAR, SEA, COCOON, SWAT, ICALP, STACS, SODA
2025 ESA (TRACK A)
2024 RANDOM, ISAAC, SOSA
2020 ICDCN
2024 – 25 **Guest journal editor** THEORY OF COMPUTING (*Special issue on RANDOM 2024*)
2024 – **Co-editor** ALGORITHMS COLUMN FOR THE BULLETIN OF EATCS (with Thomas Erlebach)

- 2015–2016 **Graduate Student Representative,**
UNIVERSITY OF MARYLAND, Computer Science Department Council
- 2012–2014 **Graduate Student Representative,**
UNIVERSITY OF MARYLAND, Computer Science Department Education Committee